

Joshua Oti

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EDUCATION

University of Southern California Master of Science, Computer Science; GPA: 3.76/4.0 • Relevant Courses: Machine Learning, NLP, Algorithms, Database Systems, Software Engineering	Los Angeles, CA May 2026
Covenant University Bachelor of Engineering, Mechanical Engineering; GPA: 4.85/5.0 • Relevant Courses: Advanced Calculus, Linear Algebra, Probability & Statistics, Numerical Methods • Academic Honors: Top 1% of Graduating Class (Mechanical Engineering); 3rd Place, ACPC (Regional ICPC)	Ogun, Nigeria Oct 2021

TECHNICAL SKILLS

• Languages: Python, C, C++, TypeScript, SQL, Java, C#, bash, Go
• AI/ML: PyTorch, MCP, LangGraph, RAGAS, LoRA, Transformers, Scikit-learn, ChromaDB, Google AI Studio
• Tools: Git, RabbitMQ, Kafka, MLflow, Figma, Playwright, GraphQL, Grafana, Prometheus, Postman, JMeter
• Cloud/Infra: Amazon Web Services, Kubernetes, Microsoft Azure, Google Cloud, Docker, CI/CD
• Certifications: AWS Certified Solutions Architect Associate (AWS CSAA)
• Web/Backend: FastAPI, REST APIs, React, gRPC, Node.js, Spring Boot, ASP.NET Core, Django, OAuth 2.0, J2EE
• Affiliations: National Society of Black Engineers (NSBE), CodePath, Association for Computing Machinery

EXPERIENCE

Autodesk Machine Learning Engineer Intern • Owned the end-to-end lifecycle of an enterprise GenAI content platform using a FastAPI backend, Azure OpenAI, and AWS S3, automating workflows to save 10+ hours weekly. • Designed dynamic LLM pipelines incorporating few-shot prompting strategies to significantly optimize generative model reasoning and image output quality. • Engineered a responsive Vue.js gamified frontend with enterprise SSO integration, driving a 20% increase in user participation through seamless, multi-device accessibility.	San Francisco, CA May 2025 - Aug 2025
Phasecurve Technologies Ltd Software Engineer • Architected a cloud-based ETL and analytics backend using Azure SQL and Flask microservices on Azure App Services, implementing autoscaling to absorb sudden 3x traffic spikes on a fintech platform. • Engineered event-driven data pipelines using Azure Serverless Functions and Azure Queues for asynchronous social media data extraction, implementing exponential backoff to reliably handle strict third-party API rate limits. • Automated CI/CD pipelines leveraging GitHub Actions and Docker, containerizing services to ensure reliable deployments and cutting deployment time by 50%. • Optimized content uploads with multi-threading and asynchronous processing, reducing backend response latency by 80%, and integrated Azure Application Insights for telemetry monitoring to rapidly diagnose pipeline failures. • Led the migration from Mailgun API to Gmail API, optimizing the email routing service and significantly reducing operational costs.	Lagos, Nigeria Nov 2022 - Sep 2024
ReHboX Machine Learning Engineer • Developed and deployed a real-time computer vision algorithm for edge devices to analyze human pose from camera inputs utilizing Python and OpenCV. • Built a predictive machine learning model using Scikit-learn alongside an interactive data dashboard, providing actionable insights on the Total Addressable Market (TAM). • Coauthored 6 technical design documents, aligning engineering workflows with user stories to streamline development sprints and cut meeting times by 15%.	Lagos, Nigeria Jun 2021 - Feb 2022

SELECTED PROJECTS

LLM Fine-Tuning & Bias Analysis PyTorch, Transformers — NLP Research • Benchmarked and optimized decoder (Llama 3.2-3B) and encoder (RoBERTa/DeBERTa) architectures to quantify and evaluate trade-offs between sentiment accuracy and social safety. • Engineered custom LoRA adapters targeting 7 projection layers with 0.18% trainable parameters, enabling highly efficient fine-tuning on A100 GPUs. • Evaluated language model stability and bias shifts using StereoSet metrics (ICAT, LMS, SS) across multiple in-context learning techniques, discovering that 8-shot prompting optimally maximized safety scores.	Nov 2025
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- Architected a comprehensive portfolio of machine learning solutions across diverse problem spaces, spanning the full ML lifecycle from foundational statistical models to advanced deep learning architectures.
- Established robust predictive baselines and feature extraction frameworks, implementing regularized linear models (Logistic, LASSO, Ridge) and time-domain processing for multi-sensor data.
- Overcame class imbalances and complex distributions in specialized datasets by engineering ensemble systems (Random Forests, XGBoost) with SMOTE oversampling and cost-complexity pruning to prevent overfitting.
- Built active learning and semi-supervised self-training pipelines using L1-penalized SVMs and Monte Carlo simulations, benchmarking label efficiency against unsupervised Spectral Clustering.
- Applied transfer learning to pre-trained vision models (EfficientNetV2, ResNet-50), optimizing feature extraction to achieve high classification accuracy on image datasets.

- Engineered an agentic web application enabling natural-language querying, automated tool routing, and interactive data analysis over a large-scale movie dataset.
- Architected a hybrid RAG pipeline combining SQL (SQLite) and vector retrieval (ChromaDB), orchestrated via LangGraph and MiniMax-m3 for stateful workflow management and intent routing.
- Leveraged Model Context Protocol (MCP) tool interfaces to enable safe, programmatic database execution and automated self-correction on complex queries.
- Integrated LangSmith for deep observability into the agent’s execution traces, enabling rapid debugging of complex tool-routing paths and monitoring of token usage.
- Validated agent responses using the RAGAS framework to measure answer quality, mitigate model hallucinations, and ensure factual accuracy.

LEADERSHIP

- **National Society of Black Engineers, Graduate Student Liaison USC Chapter**
- Apr 2025 - Apr 2026
- Bridged the undergraduate-graduate gap by organizing joint social initiatives, driving a 25% increase in graduate student event attendance.